

V.H.F. F.E.T. PREAMPLIFIER WITH 0.3 dB NOISE FIGURE

Indexing terms: Field-effect transistors, Noise, Preamplifiers

Using a Plessey gallium-arsenide f.e.t., a preamplifier having a noise figure of 0.3 dB and a stage gain of 22 dB at 129 MHz has been constructed. A summary is given of the design, and the method of alignment, involving a simple inductive 4-port bridge, is also discussed.

As the mobility of the carrier in a gallium-arsenide semiconductor is several times that of its silicon counterpart, it might be expected that the noise performance of a GaAs f.e.t. at very high frequencies would be superior to that obtained from silicon devices. Further, there exists the possibility of cooling a GaAs f.e.t. in liquid nitrogen (77 K), without the onset of 'carrier freezeout', to improve further the noise performance. To test these hypotheses, a preamplifier for use with a nuclear magnetic resonance (n.m.r.) spectrometer at 129 MHz was constructed using Plessey f.e.t.s (type GAT 1). The operating conditions met in this field of research create severe difficulties in the design of a stable amplifier; the signal source is a tuned circuit ($Q \approx 200$) whose impedance may only be transformed to, say, 50 Ω resistive at one frequency, and, further, the microvolt signal induced in that source is the transient response of the nuclear system to a pulse of 50 W (instantaneous) of r.f. power lasting say 40 μ s, applied to the same tuned circuit. Finally, the signal must be observable microseconds after the end of the r.f. pulse. It follows that an amplifier whose performance meets these demands is suitable for most applications.

Preliminary experiments indicated that a noise figure of less than 1 dB was obtainable at room temperature, and, at 77 K, the corresponding figure was less than 0.5 dB. A target of 0.2 dB was therefore selected as a design basis for an amplifier working in the common-source mode, with a drain current of between 5 and 10 mA and drain voltage of +5 V. For a noise figure of 0.2 dB to be obtainable, the drain load must contribute remarkably little noise. One method by which this may be accomplished is to make the load reactive. However, the stability of the amplifier is then poor when the gate load is tuned, for the gate-drain capacitance of the GAT 1 is approximately 0.2 pF at 77 K, and, further, the gain with reactive loading is likely to be low. A low Q -factor tuned load with neutralisation of the gate-drain capacitance was therefore chosen, and the load was capacitively matched to 50 Ω .

Neutralisation was achieved with the aid of an inverting transformer, which comprised the inductive part of the load, and a 0.5 pF variable capacitor (Fig. 1). With such an arrangement, it was found that the optimum signal source impedance was about 800 Ω resistive and the 50 Ω signal source used was transformed to this value with the aid of a tuned circuit with tapped inductance. It may be shown that, for a transconductance of 10 mA/V, the drain load resistance at 77 K contributes negligible noise if, for the source impedance stated, it is very much greater than 80 Ω . A value of 680 Ω was therefore chosen, giving a power gain for the first stage of about 22 dB into 50 Ω , and a bandwidth of about 7 MHz. Removal of the signal source resulted inevitably in oscillation as the Q -factor of the undamped gate circuit (at 77 K) was in the region of 600.

The second stage of amplification may conveniently be at room temperature, and, as the drain load of the first stage is transformed (on resonance) to 50 Ω , the 2nd-stage design may be similar to that of the first. However, the gate-drain capacitance of the GAT 1 at room temperature is considerably less than at 77 K, and so no neutralisation is needed. The optimum source impedance at room temperature is about 680 Ω resistive, and it may be shown that the second stage must have a noise figure of less than 1.6 dB if it is not to degrade the performance. Accordingly, the second stage was built at room temperature in a milled-aluminium box on top of a small Dewar vessel, while the first stage sat in a small enclosure in the liquid nitrogen, suspended from above by three stainless-steel tubes which carried, respectively, the input cable, the output cable to the second stage and the power lines. The construction of both stages aimed to shield as far as possible the drain circuitry from that of the gate, and, to this end, the first stage was built on a 5 $\times$ 5 cm copper plate which acted both as ground plane and shield. The drain and gate circuits were situated on opposite sides of the plate and at opposite ends, the transistor's being placed in the middle of one edge of the plate. The latter bisected the bottom of the f.e.t. and so shielded the gate lead from the drain lead. The source and shield were immediately grounded. It was found that most types of capacitor usually used at these frequencies functioned in liquid nitrogen provided that their values were less than about 150 pF; larger values, presumably because of the thinness of the dielectric used, tended sooner or later to malfunction. All the 100 pF capacitors used had a polystyrene

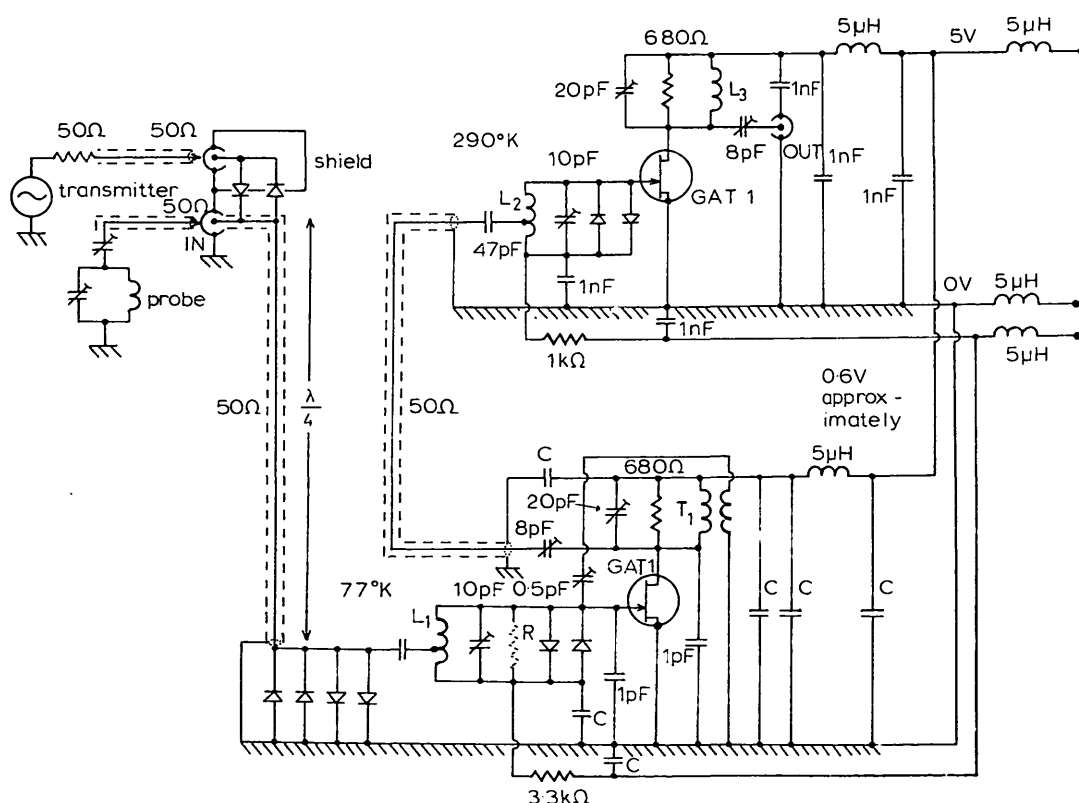


Fig. 1 129 MHz low-noise preamplifier

L_1 , 5 turns, L_2 , 6 turns 18 s.w.g. copper wire, 7 mm diameter
 L_3 , 7 turns, T_1 , 2 $\times$ 7 turns, 22 s.w.g. copper wire, 5 mm diameter

All inductors are self supporting
All capacitors C are 100 pF polystyrene
Resistor R , used only in alignment, is 800 Ω carbon
All diodes are 1N4154

dielectric, and the variable capacitors on the drain were Mullard trimmers type 808 and the gate variable capacitor was a Tekelec-Airtronic type AT5751. All resistors were 0.125 W carbon, as the latter changes its resistivity by only about 15% at 77 K, as compared with room temperature. The 0.5 pF variable capacitor was homemade, being a fixed nut and movable bolt for one plate and a small piece of copper foil for the other; the small fixed capacitors, designed to prevent gigahertz oscillation, were made from copper foil and Teflon tape glued to the ground plane.

In aligning the amplifier, much use was made of a simple 4-port inductive bridge. When r.f. is applied at the first port, there is zero output from the third port when the loading on the second and fourth ports is equal. The loading on the fourth port was always 50 Ω resistive, and the bridge could therefore be used to match circuitry to 50 Ω . Thus, by temporarily connecting 800 Ω across the gate tuned circuit, the correct tuning and tapping point on the inductance could easily be found in order to accurately transform the source impedance to the desired value. Similarly, the drain-circuit capacitors could also be accurately set, and the neutralisation capacitor was considered to be correctly adjusted when loading and unloading the signal input to the gate with 50 Ω gave no change in the drain-circuit matching. This procedure was repeated with the second stage and the two stages then connected together, remembering to remove the 800 Ω resistance from the gate circuit. It is perhaps worth pointing out that the above procedure optimises both stages for signal/noise ratio and that maximum gain occurs at a slightly different frequency. Minor variations of the neutralisation capacitor and the 2nd-stage gate capacitor were finally made to obtain the best performance, and the figure eventually reached, as measured with the aid of a hot/cold noise source, a

calibrated attenuator and an n.m.r. spectrometer was a little under 0.3 dB. Typically, under 0.5 dB was easily attained; the last 0.2 dB required patience. Even when the tuned n.m.r. source impedance deviated considerably from 50 Ω resistive at the spot frequency of interest, the amplifier did not oscillate, and only when there was a gross mismatch which had little resistance did instability commence. This situation should never occur in most fields of interest, e.g. with an aerial connected to the input, and it is perfectly possible with purely resistive signal sources to dispense with the neutralisation; the result is a detuning in opposite directions of the gate and source circuits and a reduction of bandwidth. Finally, it may be noted that all time constants associated with the circuitry are under a microsecond, and that the first f.e.t. is protected from the r.f. pulses with the aid of crossed diodes and a $\lambda/4$ line,¹ so that recovery from overload is safe and rapid.

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NEW CLASS OF PRIMARY-FEED ANTENNAS FOR USE WITH OFFSET PARABOLIC-REFLECTOR ANTENNAS

Indexing terms: Antenna feeders, Reflector antennas

In many practical applications the performance of an offset parabolic-reflector antenna is limited by the offset reflectors' depolarising and beam-squinting properties. The letter describes a new class of primary-feed antennas which overcome these limitations. The new primary-feed types offer a significant improvement in the crosspolar and beam-squinting properties of the offset reflector antenna, without adding significantly to the complexity or mass of the primary-feed system.

In many antenna systems where high gain and very low side-lobe radiation is required, the offset parabolic reflector offers a number of significant advantages over its axisymmetric counterparts. In particular, since the primary feed and its supporting structure need not protrude into the optical path of the incident or reflected wavefronts, spurious scattering, and the associated gain loss due to blockage effects, can be avoided. The avoidance of blockage is a major factor in achieving high overall efficiencies and low sidelobe performance.^{1,2}

For many practical applications the offset reflector configuration has a serious disadvantage, however, in that the reflector has undesirable depolarising properties when illuminated efficiently by a perfectly linearly polarised primary-feed radiation. The overall radiation field from such an antenna typically exhibits a pair of crosspolarised lobes, with peaks close to the -6 dB contour of the antenna main beam, in one of the principal planes of the reflector. In the other principal plane the crosspolarised field is zero.^{3,4} The plane containing the peak crosspolarised lobes will be referred to here as the plane of asymmetry. When similarly illuminated by a purely circularly polarised primary feed, the offset reflector does not introduce a crosspolarised field, but the main beam of the antenna is squinted from the antenna bore-sight.³⁻⁵ These depolarising and squinting phenomena are often undesirable in high-performance antennas, especially when frequency reuse is required. In many cases these effects constitute a major limitation on the application of the offset reflector, regardless of its other desirable properties.

To overcome these limitations in offset reflector antennas, without adding excessively to the complexity or mass of the

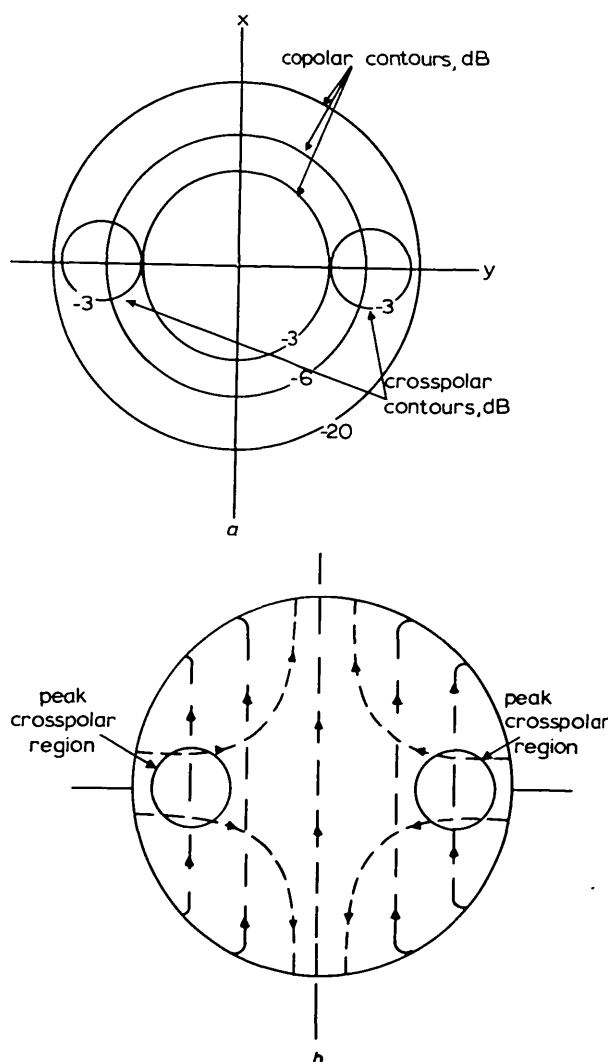


Fig. 1

a Contour plot of typical focal-plane field characteristics for an offset parabolic reflector. Crosspolarised field peak level typically -20 to -30 dB below peak copolarised field
b Aperture-field distribution of overmoded waveguide horn